

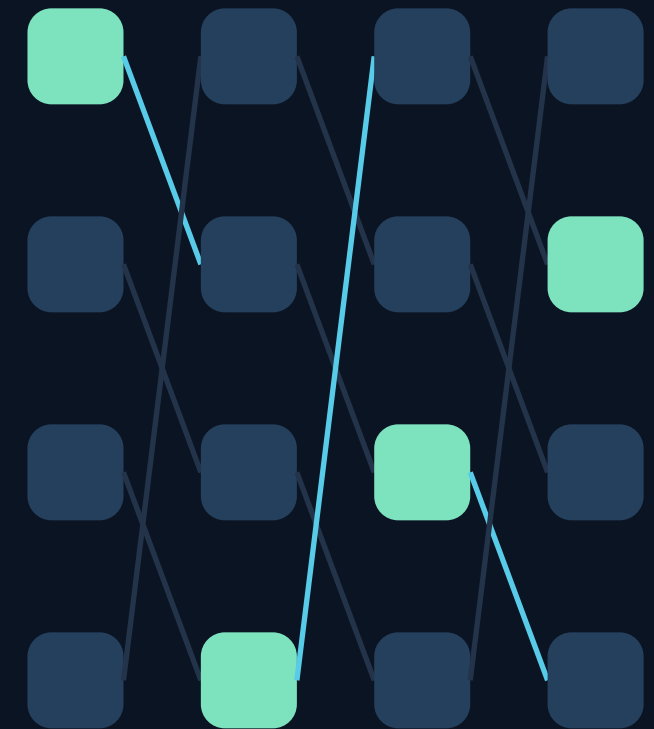
Sleeping Machines

A trainable substrate for intelligence through time and sparse events.

Build useful capacity faster than active
computation.
From datacenters to mobile devices and robotics.

Tero Keski-Valkama | Sole founder

Proposed raise €3M · Bullish case €50M pre-money
Research-stage deep tech · 3 October 2026



Capacity beyond activity

A new model family. A new computing substrate.

The goal: useful intelligence with less active work, persistent state and adaptation.

CAPACITY BEYOND ACTIVITY

More memory

Grow available learned state while selecting a small amount of computation for each event.

TIME COMPUTES

Learned races

Message content, delays and local state determine which computation happens and when.

CLOUD TO EMBODIMENT

One substrate

Pursue models and runtimes first; extend the same principles to edge systems and future clockless silicon.

The investment thesis is an architecture-runtime-hardware platform, with staged proof.

The constraints reach from power grids to robot batteries.

Economic context for efficient intelligence; these figures are not our revenue market.

DATACENTERS

485 → 950

TWh/year, 2025 → 2030 central IEA projection. Power and data movement make system efficiency economically important.

MOBILE

5.8B

Unique mobile subscribers in GSMA's 2026 report. On-device inference creates a prospective route to broad distribution.

ROBOTICS

5M

Industrial robots operating in 2025. Sparse sensing, persistent state and adaptation are prospective use cases.

Select one measurable initial workload; expand only after quality and system gains hold.

Hard sparse computation that can learn its alternatives.

Conceptual event path; the trained language core is implemented in a synchronous software emulator.

01 / MESSAGE

Content + time

Encode incoming signal. Content and persistent memory influence the emission delay.



02 / RACE

Hard winner

Score candidate keys. A temporal race selects an addressed receiver.



03 / MEMORY

Local update

Deliver the selected value and evolve local persistent state through deep layers.



04 / LEARNING

Route credit

Train contents and timings using selected and unrealized alternatives; charge the credit work.

Counterfactual credit: learn from selected and unrealized routes

Inference selects values sparsely; current training still computes losing alternatives.

Three results make the architecture worth scaling.

Completed text8 experiments; all numbers remain exploratory single-seed evidence.

LANGUAGE QUALITY

2.162 bpc

Best p96 / D4 result is slightly ahead of saved one-pass LSTM at 2.171; it uses more work.

CAUSAL MECHANISM ABLATION

−0.135 bpc

Adding value-based route credit improves the same p32 / D4 model with +0.30% estimated fitting work.

CAPACITY / SELECTION

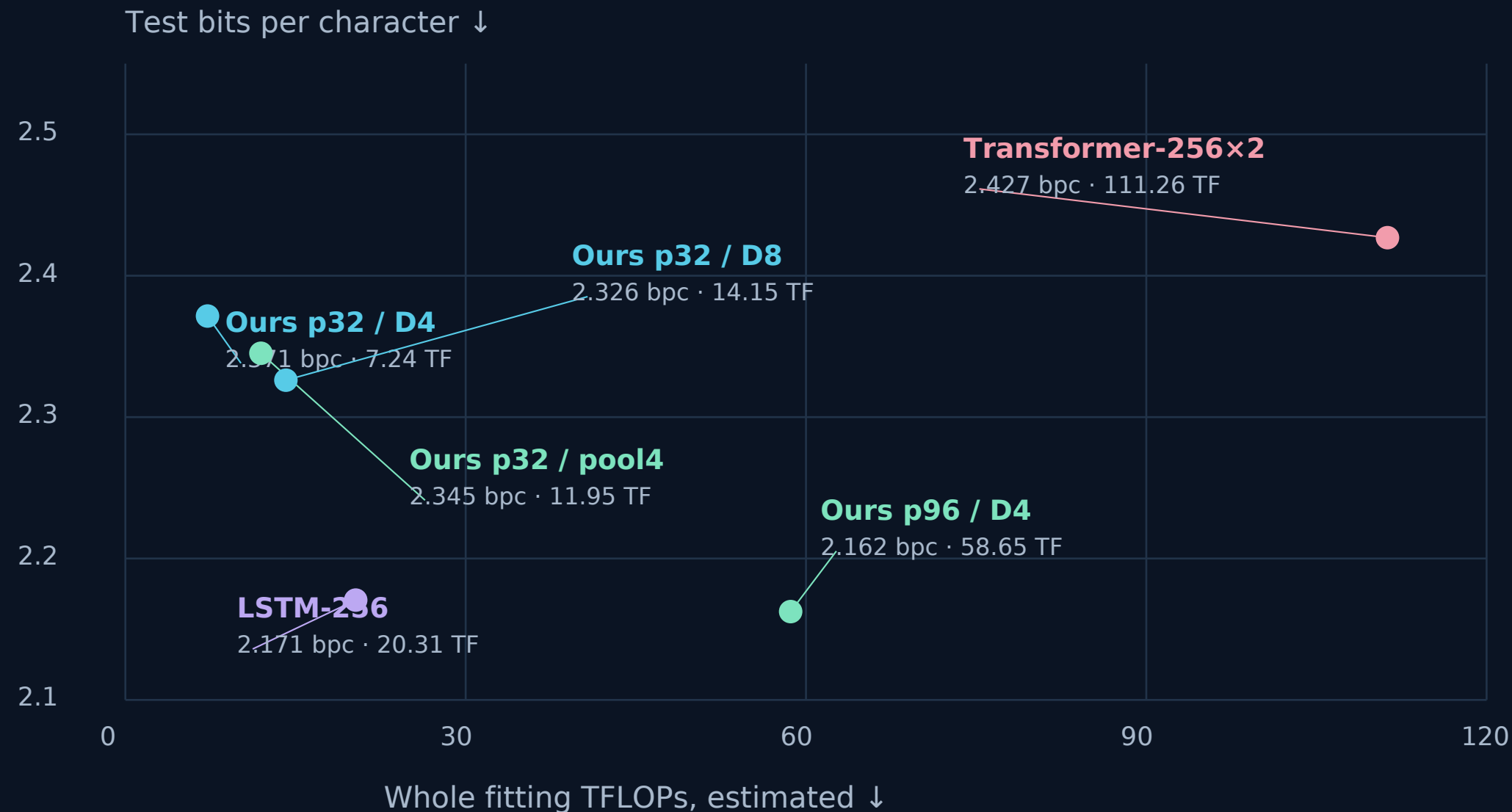
8 writes

Doubling the credited p32 memory pool improves quality while keeping selected writes per input position fixed.

Distinctive mechanisms affect learning; scale, modern controls and serving economics remain open.

A promising quality-work frontier at small scale.

Same text8 held-out region and T256 windows; saved one-pass controls, not optimized frontier baselines.



QUALITY FRONTIER

2.162

Ahead of saved LSTM; 2.89× fitting work.

SMALL TRANSFORMER CONTROL

15.4×

p32 / D4: less estimated fitting work and better quality.

BPC ↓ is better. Fitting includes credit/backward/clip/Adam estimates; conventions and training order differ. One seed. No wall-time or energy claim.

Learning the alternatives changes the result.

The same p32 / D4 model, data budget, seed and evaluation window.

TEST BITS / CHARACTER

2.506 → 2.371

Timing-only credit versus added local alternative-value route credit.

ESTIMATED FITTING WORK

+0.30%

Hard forward selection and parameter count retained.

CHARACTER PERPLEXITY

−8.9%

Derived from the BPC difference; one controlled seed.

Same p32 / D4 model; hard forward routes retained

Depth-8 also improves by 0.130 bpc with route credit. This supports the credit mechanism across two depths; replication is next.

A falsifiable architectural mechanism, with a small counted arithmetic increment.

More available state. The same eight selected writes.

Matched credited p32 / D4 pool expansion: 16 → 32 slots.

Pool 2



16 slots · 512 value scalars
8 writes · 16 keys scored
2.371 bpc

Pool 4



32 slots · 1,024 value scalars
8 writes · 32 keys scored
2.345 bpc

WHAT SCALES

8 writes

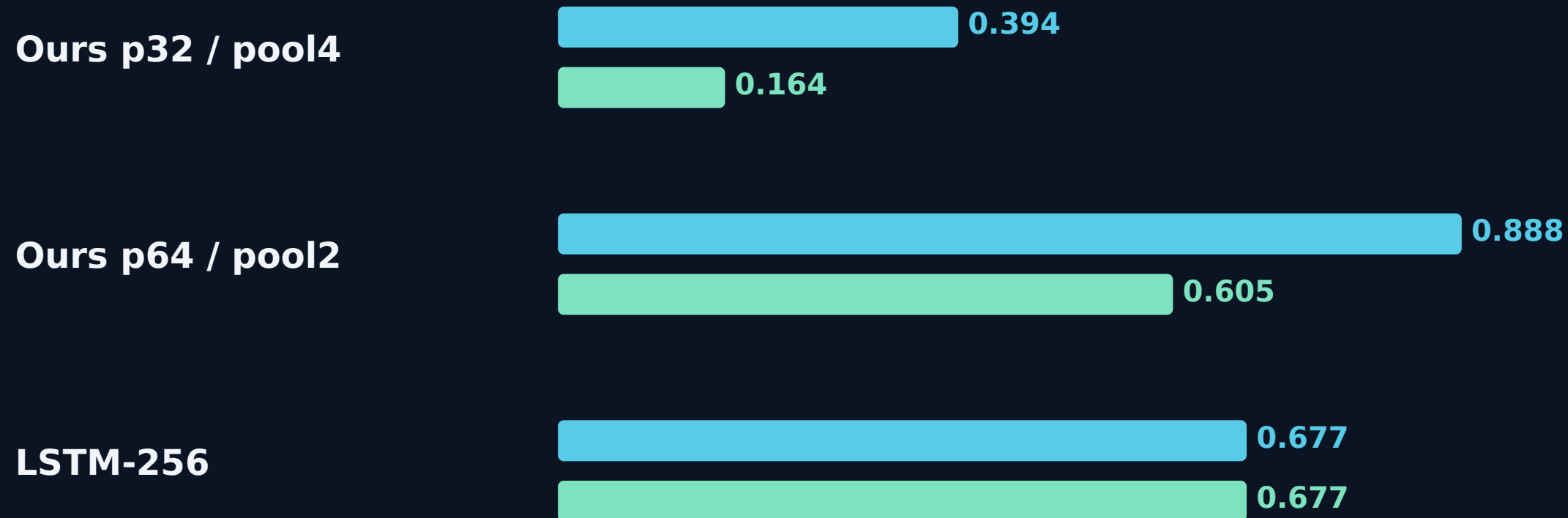
Winner-only arithmetic +0.66%.
Fitting work +65.0%. Keys and
value storage double.

Quality improves by 0.026 bpc; total capacity and learning costs remain explicitly charged.

The sparse path is implemented; system proof is next.

Traced arithmetic per input position. Only selected proposal values are computed in the winner path.

Estimated MFLOPs per evaluated input position



STATUS

All keys remain scored. Actual trained FP32 parity and held-out rescore are pending. Traffic, setup, latency and joules must be measured.

Cyan: emulator / control forward Green: winner-only / same control

These are backend/shape estimates. Native evaluation pays overlapping-window warmup (~2 positions/target); LSTM recurrent evaluation differs. Not a service-cost win.

An adaptation signal exists. Its work is accounted for.

An earlier integrated CPU model, on a new development stream; separate from the latest language fits.

PREDICT BEFORE UPDATE

3.191 → 3.096

8,191 new development targets; same persistent state, frozen versus full-backbone adaptation.

COSTED LEARNING

10.7× work

Total online processing work includes credit, backward and optimizer. Retention under drift and native asynchronous learning remain open.

Online-learning upside: persistent adaptive systems. Current result: one costed pilot.

The contribution is the complete construction.

Established alternatives already address parts of this problem; they remain essential controls.

Approach	Established strength	Our comparison obligation
Transformer / MoE	Strong quality; MoE separates available parameters from selected expert work.	Match quality/data and charge routing, memory traffic and training.
RNN / selective SSM	Persistent state and efficient sequence processing.	Test whether temporal races add useful quality or allocation.
Neuromorphic systems	Event-driven hardware and local learning precedents.	Prove expressive model quality, mapping and measured energy.
Sleeping Machines	Joint timing, sparse memory, deep representation and alternative credit.	Scale integrated mechanisms without replacing them with dense carriers.

Sparsity, recurrence and on-chip learning alone are not the novelty claim.

Enter through software. Earn the hardware expansion.

Proposed commercialization sequence; no customers or partner interest yet.

01 / MODEL + RUNTIME

First wedge

License a validated model/runtime for a chosen persistent sequence workload. Demonstrate quality and complete serving cost.

02 / ACCELERATOR CO-DESIGN

Partner path

Port verified kernels, profile memory and event execution, then pursue a joint platform or IP license.

03 / CLOCKLESS SUBSTRATE

Long-term upside

Explore delay precision, sparse local state and event scheduling in FPGA/ASIC feasibility work after software gates pass.

The €3M round funds evidence and feasibility; a full advanced-node tapeout is outside this budget.

Incumbents have reasons to evaluate a new substrate.

Potential counterparties only. Strategic relevance is an inference; no relationship is claimed.

AMD

Models + silicon

Silo AI acquisition and Liquid AI collaboration show appetite for model/software capabilities. Earn attention with AMD-compatible scale and measured gains.

INTEL

Event systems

Hala Point and Loihi 2 establish event-computing research. Offer a credible expressive model and hardware mapping experiment.

GOOGLE

Full-stack co-design

TPU 8t/8i embody model-hardware optimization. Offer independently validated improvements that complement a large existing research stack.

A measured advantage could become strategic leverage against incumbent economics; interest must be established.

AMD's Silo AI deal supports the strategic logic.

Announced approximately \$665M all-cash acquisition in 2024; a mature-company precedent.

WHAT AMD ACQUIRED

Team + stack

AI scientists and engineers, enterprise solutions, model development and software capability on AMD hardware.

THE LUMI CONNECTION

Pororo / Viking

AMD highlighted multilingual models and training on AMD-powered LUMI. LUMI is the supercomputer, not the model name.

OUR LESSON

Prove leverage

Build a capability that improves a platform's economics and adoption. One small research project is not a direct pricing comparable.

Efficient models can matter to a chip ecosystem; training on its chips is one part of the value case.

Capture a fraction of verified value created.

Illustrative annual economics, contingent on quality and complete system savings.

Scenario input	Arithmetic	Potential annual revenue
Datacenter license	€1B eligible spend × 20% savings × 20% capture	€40M
Mobile deployment	100M licensed new devices × €0.50 royalty	€50M

Datacenter customer retains €160M of the assumed €200M gross savings before migration and other costs. Robot fleets could support per-system adaptation licenses after validation.

These are assumptions for value-capture design, not booked revenue, forecasts or an additive TAM.

A founder at the intersection of models and systems.

Public professional record and a primary published invention document.

Tero
Keski-Valkama

Sole founder

Public work spans machine learning, software architecture and industrial applications. The relevant founder case is sustained technical execution across these layers.

2012-2018

Cybercom · Software Architect

Historical role in public professional profile.

2018-2022

HERE · Lead ML & AI Engineer

Self-reported research and applied ML leadership.

2022-2025

Alloy.ai → Kaiko.ai

Staff engineering; multimodal foundation-model work.

2023 publication

EP4148389A2 · Named inventor

Maps from geospatial observations; applicant HERE.

Career dates are self-reported. No PhD, current employer or ownership of HERE inventions is asserted. Sole-founder status supplied by Tero.

Build a moat around evidence, execution and integration.

The repository is now private; future progress disclosure needs a deliberate policy.

RESEARCH + DATA ROOM

Auditable

Versioned results, ablations, numerical contracts and retained negative findings support credible technical diligence.

RUNTIME + CO-DESIGN

Compound

Model, credit estimator, sparse execution, measurement and hardware mapping can become mutually reinforcing know-how.

IP + ADOPTION

To establish

Clarify contribution rights, licensing and employer assignments; then build deployment tooling and partner-specific integration.

Private source control helps disclosure discipline; exclusive IP and adoption still require proof.

Use capital to retire the decisive uncertainties.

Milestones are proposed gates, not completed deliverables or guaranteed dates.

MONTHS 0-3

Reproduce

Hire initial team. Admit trained sparse parity, complete 90M evidence and three-seed comparisons. Resolve IP and data-room rights.

MONTHS 3-6

Measure

Benchmark latency, traffic, memory and joules on one chosen workload with tuned controls. Target $\geq 20\%$ full-system cost reduction at matched quality.

MONTHS 6-12

Demonstrate

Pursue independent reproduction and prospective design-partner trials. Validate willingness to pay and a deployable runtime.

MONTHS 12-18

Expand

Scale a winning workload; test drift/retention and a hardware feasibility path. Raise the next round only on stronger evidence.

If a gate fails, diagnose or narrow the wedge before increasing compute or committing to silicon.

€3M to scale research and engineering aggressively.

Proposed 18-month budget; assumptions to refine with hiring and compute quotes.



- Team: research + systems **€1.35M | 45.0%**
- Compute + independent replication **€0.90M | 30.0%**
- Hardware feasibility + measurement **€0.25M | 8.3%**
- IP, legal, operations **€0.20M | 6.7%**
- Contingency reserve **€0.30M | 10.0%**

€2.70M operating plan + €0.30M reserve · Average planned operating burn €150k/month.
Team assumption: average six FTE including founder at €150k/year fully loaded. No full advanced-node tapeout allocation.

Price the platform option with explicit assumptions.

€50M is a bullish negotiating case; €100M is a stretch case requiring substantially stronger conviction.

BULLISH PRICED ROUND

€50M

€3M raise → €53M post-money;
5.66% new-investor ownership.

REVERSE UNDERWRITING

6.74%

Required platform-success
probability in the stated €10B exit
scenario.

STRETCH SCENARIO

€100M

€100M pre-money requires 13.49%
in the same model; 2.91%
ownership.

$$V = p \times \text{exit equity} \times \text{future retention} / (1 + \text{discount rate})^{\text{years}}$$

Assumptions: €10B exit; 30% stake retention; 10 years; 15% discount; failure value zero.

Required odds are reverse-calculated conditions. The benchmarks do not estimate those probabilities.

Successful platform adoption can support very large outcomes.

Conditional commercial arithmetic illustrates the scale required; no forecast or market multiple claim.

Successful outcome assumption	Value capture assumption	Equity-value scenario
€10B/year eligible compute spend	20% savings × 20% capture = €400M/year; 25× revenue assumption	€10B
€50B/year eligible compute spend	20% savings × 25% capture = €2.5B/year; 20× revenue assumption	€50B

A cross-platform substrate could participate in cloud inference, persistent agents and embodied systems. Reaching these outcomes requires sustained quality, scale, margins and broad adoption.

The upside is large if the platform works and wins adoption. Its probability remains unmeasured.

Fund the evidence that makes a new substrate investable.

Private discussion draft · €50M pre-money bullish case · No customer interest reported.

€3 million

Scale the integrated architecture.
Prove the economics. Build the platform.

01

Replicate quality and counterfactual learning at meaningful scale.

02

Demonstrate sparse serving gains with complete resource boundaries.

03

Turn verified advantages into partnerships and a hardware roadmap.

Tero Keski-Valkama · Private investor diligence available by arrangement

Quality and work, in the same units for every model.

Completed one-pass 10M text8 fits; T256 held-out evaluation; estimated operation counts.

Model	Test bpc	Params K	Fit total TF	Fit MF/pos	Emul MF/pos	Winner MF/pos
p32 / D4 timing-only	2.506	108.9	7.22	0.722	0.240	0.163
p32 / D4 credited	2.371	108.9	7.24	0.725	0.240	0.163
p32 / D4 pool4 credited	2.345	177.0	11.95	1.196	0.394	0.164
p32 / D8 credited	2.326	210.0	14.15	1.416	0.463	0.310
p64 / D4 credited	2.183	422.5	26.79	2.680	0.888	0.605
p64 / D4 pool4 credited	2.179	689.8	44.07	4.410	1.458	0.607
p96 / D4 credited	2.162	940.9	58.65	5.868	1.947	1.324
LSTM-256	2.171	338.4	20.31	2.032	0.677	0.677
Transformer-256x2	2.427	1658.9	111.26	11.133	3.710	3.710

Fit MF/pos = per training position. Inference MF/pos = per evaluated input position. Native quality uses training evaluator; winner-only trained parity/rescore pending. Native overlap warmup adds work per scored target. Full 15-row CSV retains older models.

The comparison boundaries are part of the evidence.

Maintain the original records and disclose what a result actually measures.

Boundary	Current completed evidence	Remaining work
Data / evaluation	Native: text8[0:10M], random 64×128 segments, ~9.994M positions; test[95M:96M], T256.	Tail target counts and training order differ from E64; align modern controls.
Learning / seeds	1,220 native updates (latest rows), lr .004 cosine, one seed; state reset per fit segment.	Replication, robust recipe, depth/horizon and 90M completed held-out results.
Sparse execution	All keys scored; selected values only in winner evaluator; small random FP64 equality fixtures.	Actual trained FP32 winner/state/cache equality and held-out rescore.
System economics	Arithmetic traces/formulas; persistent local state within windows.	Setup, warmup, traffic, memory, throughput, latency, joules and total cost.

Keep the failures beside the positive mechanism results.

A disciplined research program should revise interpretations when evidence changes.

Finding	Interpretation	Response
Uncredited D8 pool expansion worsens quality	More capacity needs learnable selection; it is not automatically useful.	Retain negative pair; test credited expansion with matched depth/seed.
Write-credit variants regress or diverge	A particular local estimator is unstable; not all route credit is equivalent.	Withdraw variants; preserve original result and derive corrected credit.
Best p96 costs more than LSTM	Small quality lead is not comparable-quality resource superiority.	Tune and scale; require complete quality-cost frontier.
Event-vision screening is mixed	Small coarsening gains do not clear every full-screen seed or strong RBF controls.	Avoid cross-domain supremacy claim; keep modality-specific gates.

The price depends on dilution, exit scale and time.

Required platform-success odds at 10 years and 15% discount; zero failure value.

Exit equity	Stake retention	Required p: €50M	Required p: €100M
€3B	10%	67.43%	134.85%
€3B	30%	22.48%	44.95%
€3B	50%	13.49%	26.97%
€10B	10%	20.23%	40.46%
€10B	30%	6.74%	13.49%
€10B	50%	4.05%	8.09%
€30B	10%	6.74%	13.49%
€30B	30%	2.25%	4.50%
€30B	50%	1.35%	2.70%

Sensitivity analysis, not observed probabilities. No preferred-stock waterfall, intermediate distributions, follow-on investment cash flows or option-pool adjustment modeled.

A private evidence pack and primary external context.

Source labels in slide footers are clickable; companion files must travel with the PDF.

R1 **Private completed language evidence ledger (3 October 2026)**

Saved native fits and E64 controls. Original JSON records and SHA256 hashes are supplied in the private diligence pack.

R2 **Private online-learning experiment, 8,191 targets**

Predict-before-update pilot: quality and complete processing-operation estimates; one checkpoint, stream and learning rate.

R3 **Architecture, numerical contracts and report**

Private technical report. Temporal races, counterfactual credit and native sparse inference; trained-weight sparse parity remains pending.

M1 **IEA — Key Questions on Energy and AI, executive summary**

485 TWh datacenter electricity in 2025; central projection 950 TWh in 2030. Sector context, not addressable revenue.

M2 **GSMA — The Mobile Economy 2026**

5.8 billion unique mobile subscribers and 8.8 billion connections. Subscribers are not annual device sales or customers.

M3 **IFR — World Robotics 2026 release, 24 September 2026**

5 million operating industrial robots in 2025 and more than 600,000 annual installations; no adoption by this project.

T2 **Fedus et al. — Switch Transformers**

Capacity beyond selected computation has established MoE precedent; sparsity alone is not a novelty claim.

Public precedents establish relevance, not endorsement.

Historical roles are self-reported; inventorship is checked against the primary published document.

S1 **AMD — Silo AI acquisition completion, 12 August 2024**

Announced approximately \$665M all-cash acquisition; Poro and Viking on AMD. Mature-team strategic precedent, not a pre-seed valuation comparable.

S2 **AMD — Acquisition announcement and LUMI collaboration, 10 July 2024**

AMD highlighted enterprise solutions, expertise, software and Poro/Viking training on LUMI using AMD hardware; no single exclusive acquisition cause established.

S3 **Liquid AI — \$250M Series A announcement, 13 December 2024**

Financing announcement and AMD collaboration on efficient models; funding amount is not a disclosed company valuation.

S4 **AMD Ventures — investment focus**

AI models, infrastructure and silicon investment focus; published Series A-C focus means this project may be earlier than typical admission.

S5 **Intel — Hala Point research system, 17 April 2024**

Loihi 2 neuromorphic research prototype, event computing and prospective continuous learning. No Intel interest in this project claimed.

S6 **Google — Eighth-generation TPU architecture announcement**

TPU 8t/8i, model-hardware co-design and sparse MoE infrastructure. Existing internal capability creates both potential fit and competition.

F2 **European Patent Office — published application EP4148389A2**

Primary published document names Tero Juhani Keski-Valkama as inventor and HERE Global B.V. as applicant. Experience evidence, not Sleeping Machines-owned IP or proof of a grant.